

ALGORITHM AND DATASET

Student Performance Analytics

1. Algorithm

The Student Performance Analytics system analyzes student academic data such as attendance, assignment marks, internal marks, study hours, previous marks, and final marks. The data is cleaned and processed before analysis. The system calculates performance values and can classify students as Excellent, Good, Average, or Needs Improvement.

2. Step-by-Step Procedure

Step 1: Collect student details, marks, attendance, assignments, and study information.

Step 2: Store the data in a CSV file, Excel file, or database.

Step 3: Clean the data by checking missing, duplicate, and incorrect records.

Step 4: Select useful features such as attendance, assignment marks, internal marks, study hours, and previous marks.

Step 5: Calculate average marks, percentage, grades, and subject-wise performance.

Step 6: Divide the data into training and testing data if machine learning is used.

Step 7: Train the selected machine-learning model using the training data.

Step 8: Test the model using the testing data and measure its accuracy.

Step 9: Display results using charts, graphs, tables, and a dashboard.

Step 10: Generate a report and identify students who may need academic support.

3. Logic / Method Used

The system compares different academic factors with student performance. Attendance, assignment marks, internal marks, study hours, and previous marks are used as input features. The system calculates overall performance and groups students into performance levels. For prediction, a Decision Tree Classification model can be used because it is simple to understand and suitable for this type of project.

4. Model / Technique Applied

Data Analysis: Used to calculate averages, percentages, grades, and performance trends.

Data Visualization: Used to display results using charts and graphs.

Machine Learning: Decision Tree Classification can be used to predict the student's performance category.

5. Process Flow

Student Data → Data Collection → Data Cleaning → Feature Selection → Data Analysis → Training/Testing Split → Model Training → Model Testing → Prediction → Dashboard and Report

6. Dataset

A sample student-performance dataset can be used for this project. If real institutional data is available with proper permission, the same dataset structure can be applied. A sample dataset avoids using private student information.

7. Source of Data

The data can come from a public student-performance dataset, an educational institution with permission, or a project-created sample dataset. For demonstration, a sample CSV dataset can be created.

8. Number of Records / Samples

A sample dataset of **1,000 student records** can be used. Each row represents one student and contains academic and related information.

9. Features / Attributes

Student_ID – Unique student number.

Gender – Gender of the student.

Age – Student age.

Study_Hours – Average study hours per day.

Attendance – Attendance percentage.

Assignment_Score – Assignment marks.

Internal_Marks – Internal examination marks.

Previous_Marks – Previous examination marks.

Final_Marks – Final examination marks.

Performance – Target category: Excellent, Good, Average, or Needs Improvement.

10. Training and Testing Split

The dataset can be divided into **80% training data** and **20% testing data**. For 1,000 records, approximately 800 records are used for training and 200 records for testing.

11. Data Format / Type

The dataset can be stored in **CSV format**. Numerical columns include age, study hours, attendance, marks, and scores. Categorical columns include gender and performance category. CSV is simple and can be easily processed using Python and Pandas.

12. Expected Result

The system is expected to show student performance through charts, tables, and reports. It can identify high- and low-performing students, compare attendance with marks, analyze important academic factors, and predict a performance category when the machine-learning module is used.

13. Conclusion

The proposed algorithm provides a simple flow from data collection to data cleaning, analysis, model training, testing, prediction, and reporting. The dataset contains important academic features that can be used to understand student performance. This system can help teachers identify students who need support and make better academic decisions.